

AGP: Short for Accelerated Graphics Port, this is a port made exclusively for graphics cards and is significantly faster than PCI. There are different AGP standards based on speed, such as 1.0 (1x/2x), 2.0 (2x/8x) and 3.0 (4x/8x). The AGP 1x interface operates at a clock speed of 66 MHz, AGP 2x operates at 133 MHz and so on—the multiple of the AGP device can be used to calculate the clock speed of its interface.

BIOS: The Basic Input Output System is a program located on the CMOS chip located on the motherboard. The BIOS controls the primary functions of the hardware and facilitates low-level communication between the OS and the hardware. The BIOS can be upgraded to provide additional functionality.

SDRAM: Short for Synchronous Dynamic RAM, this is a type of memory that is synchronised with the FSB of the processor. It is available in various speeds such as 66 MHz, 100 MHz and 133 MHz.

DDR RAM: Double Data Rate RAM is a type of memory that handles data at twice the speed of the older SDRAM. DDR RAM typically operates at 266 MHz, 333 MHz and 400 MHz effective speeds, while the actual speeds are 133, 166 and 200 MHz respectively.

DDR2 RAM: Double Data Rate 2 RAM is a memory type based on chips designed to run at speeds higher than that of DDR RAM, such as 533 MHz and 667 MHz. Very few motherboards right now support DDR2 RAM. In our tests this month, only the Mercury PI915GVM supported it.

Dual Channel: When two RAM modules are plugged into two slots marked for two different memory channels, then, theoretically, the data flows at twice the normal rate through the RAM. This is known as a dual-channel memory configuration.

FSB: The Front Side Bus speed is the clock speed at which different components in a computer speak to the CPU. The CPU's FSB is very important, and changing it can dramatically affect the overall performance of the computer. Increasing it can improve system performance but will also void your CPU's warranty, and may sometimes cause damage to the CPU due to excessive heat generated at higher clock speeds. Decreasing the FSB reduces performance,

but may be necessary when the highest FSB supported by the motherboard is lower than the default FSB of the CPU.

IDE: IDE stands for Integrated Drive Electronics, and is a standard interface for hard drives, optical drives and other physical storage devices. It is also known as ATA, which stands for Advanced Technology Attachment. ATA33, ATA66, ATA100 and ATA133 are its various types based on the transfer speed. For example, ATA100 means that the theoretical data transfer rate of the interface is 100 MB/s. It uses a 40 pin connector cable that may consist of 40 or 80 conductors.

Multiplier: This is the ratio between the CPU speed and the FSB speed. As an example, a CPU with an FSB of 200 MHz and a multiplier of 18 will work at $200 \times 18 = 3,600$ MHz.

Northbridge: This is a chip on the motherboard that controls the FSB and memory. This is the largest chip on the motherboard and is normally covered by a heat sink, and sometimes actively cooled by a fan.

PCI: Peripheral Component Interconnect is a standard that defines the connection between the motherboard and expansion cards. The data transfer rate of PCI bus can be up to 133 MB/s.

PCI-Express: Peripheral Component Interconnect Express is a new PCI standard that transfers data at speeds equivalent to AGP 16x, which is roughly 4.3 GB/s.

RAID: Abbreviation for Redundant Array of Independent Disks, RAID is a software or hardware configuration that employs two or more drives in combination for fault tolerance and performance.

Fault tolerance is a method used to safeguard data by writing the same data to two different partitions or different hard drives so that a redundant copy is available in case the first gets corrupted or destroyed. This involves two write operations at a time, so naturally, system performance suffers. RAID 0, 1 and 0+1 are the commonly used types of RAID.

RAID 0: This involves striping only, and there is no data redundancy. It has the best performance but no fault tolerance. Striping is a technique in which the drives' storage space is partitioned into units

ranging from 512 bytes to several megabytes. These stripes are interleaved and addressed in order.

RAID 1: This type of RAID implementation is also known as disk mirroring, which involves at least two drives to duplicate data. Striping is not involved here. This offers the best ratio of performance to fault tolerance.

RAID 0+1: This combines the best features of RAID 0 and 1. In this type of RAID, two pairs of striped drives are mirrored together to provide fault tolerance. Thus, it requires four hard drives to implement.

Matrix RAID: This is a technology developed by Intel, which achieves what RAID 0+1 achieves using only two hard drives. It does so by creating two separate volumes in an array of two hard disks. It enables you to separate the two-disk Array into two volumes, as if there were two separate sets of hard disks.

SATA: Also abbreviated as S-ATA, SATA is short for Serial ATA, which is an interface for physical storage devices such as hard drives. SATA uses a serial technology in which data is moved in a single stream, which is done much faster than a parallel interface because unlike the parallel interface, a serial interface is not tied to a particular clock speed. The transfer rates for SATA begin at 150 MBps, and for SATA II it is 300 MBps.

Southbridge: This is the chip on the motherboard that controls all the onboard devices including the IDE bus and PCI bus. This is the second-largest chip on the motherboard and is sometimes covered by a heat sink.

CNR slot: CNR is an abbreviation for Communications/Network Riser, and the CNR Slot was developed by Intel. It is a slot on the motherboard that allows audio and modem devices created for this standard to be plugged into it.

OpenGL: Short for Open Graphics Language. It is a vendor-neutral, multi-platform programming language used mainly to create 3D applications and games. It was developed by SGI.

Underclocking: Reducing the clock speed of the CPU. It is achieved by either reducing the FSB speed or the multiplier of the CPU.

The winner in this round was once again the MSI RS480M2-IL, which stamped its authority over all the other boards with a thumping 3918 overall score.

The second-fastest board, again, the TUL A480M7-VF, could manage 3408, significantly less than the MSI board.

The MSI board's score here

was the second-best of all the AMD boards and was beaten only by the WinFast NF4K8MC, which had the advantage of a 3D card.

The rear was again brought up by the MSI RS 480-M which managed a mere 2125, almost half of what the winner in this test scored!

Other Synthetic Benchmarks

We then benchmarked all these boards on our trusted SiSoft Sandra 2005 Professional and measured various sub-system performances. This test gives an individual breakup of performance of the various aspects of the system.